

**AN INTELLIGENT SYSTEM FOR DETECTION,
TRANSLATION, AND HISTORICAL MAPPING OF
ANCIENT STONE INSCRIPTIONS FROM THE
ANURADHAPURA KINGDOM SRILANKA**

25-26J-144

Project Proposal Report

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B.Sc. (Hons) in Information Technology Specializing in Data Science

Department of Computer Science -Faculty of Computing

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ABSTRACT

The goal of this study is to develop an artificial intelligence system for translation and knowledge extraction from ancient Anuradhapura Kingdom Brahmi inscriptions. The central research problem is the challenge of deciphering fragmentary, broken, and compound language Brahmi inscriptions, which makes proper and meaningful translation impossible. The goal is to transform digitized Brahmi text into proper, culturally acceptable modern translations in Sinhala and English and to extract structured knowledge of historical, cultural, and administrative context for academicians.

The project takes a multi-stage computational approach beginning with downloaded digitized texts of Brahmi from character recognition models. The algorithm incorporates linguistic information and natural language processing (NLP) such as entity recognition, topic modeling, and creation of a knowledge graph. These steps facilitate handling ambiguity and partially given texts as well as semantic extraction from translated text. Refining includes feedback loops of iterative iterations with restoration modules to increase accuracy and completeness in translations.

While the final conclusions arrive upon completion, the approach anticipates producing highly accurate bilingual translations and dense structured data sets revealing heretofore inaccessible historical trends and observations through manual means. The knowledge graphs to be derived from NLP should enable advanced querying, digital preservation, and enhanced understanding of socio-cultural and administrative details of the Anuradhapura period.

Lastly, this module combines ancient epigraphy and modern AI technologies to significantly augment the digitization, translation, and semantic analysis of Brahmi inscriptions. Its findings will enhance historical research and preservation activity by providing accessible, reliable translations and a semantic model by which to explore Sri Lanka's ancient history.

DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BCE	Before Common Era
CNN	Convolutional Neural Network
KG	Knowledge Graph
ML	Machine Learning
NLP	Natural Language Processing
NER	Named Entity Recognition
OCR	Optical Character Recognition
RDF	Resource Description Framework
GPU	Graphics Processing Unit
API	Application Programming Interface

1. INTRODUCTION

1.1 Background & Literature Survey

The Brahmi inscriptions of the Anuradhapura Kingdom, covering the 3rd century BCE to early centuries CE, are among the most valuable primary historical sources of information regarding the early Sri Lankan civilization. The inscriptions relate to a vast range of things from royal announcements, grants of land, religious donations, and administrative instructions to provide useful information on the political institutions, linguistic evolution, religious cults, and social customs of the time. Previously, epigraphists and historians have interpreted by-hand these eroded inscriptions through laborious visual examination and expert interpretation, a slow, sweat-and-tears, and error-ridden procedure vulnerable to inconsistencies resulting from damage, wear, and fragmentary texts. The script itself, an archaic type of Brahmi, is challenging due to the archaic character types, local variations, and linguistic complexity, aggravated moreover by erosion and physical deterioration of stone inscriptions over the centuries.

During recent decades, the melding of computer vision, machine learning, and natural language processing (NLP) has offered promising avenues for automatic transcription and translation of some of this work. Notably, projects such as Hettiarachchi et al.'s "Brahmi to Sinhala Translator" (2020) have demonstrated how Optical Character Recognition (OCR) technology tuned to ancient scripts may be integrated with NLP pipelines to translate images of Brahmi inscriptions into digital text in Sinhala. This approach combines image preprocessing for denoising and enhancing clarity, convolutional neural networks (CNN) to identify characters, and text mining techniques to parse and transliterate Brahmi characters accurately. These advancements reduce human effort and provide scalability to epigraphic research, even though problems persist in processing inscriptions with missing parts and ambiguous symbols.

Aside from script transcription, sophisticated NLP methods have been more employed to translate and semantically process historical text-derived content. Methods like Named Entity Recognition (NER), topic modeling, and the construction of knowledge graphs have been applied in domains like multilingual corpora of history and cultural heritage documentation to recognize, classify, and connect entities like places, people, events, and administrative names within ancient narratives (Graciotti, 2023). These semantic assets offer grades of organized information, supporting deeper querying and aiding digital preservation projects.

Furthermore, researchers point to the necessity for round-trip iterative feedback loops among restoration modules (employed to repair impaired text and images), recognition engines, and translation components to support progressive improvement based on context and partial information. This is a cycle that assists in recovering lost text data, improving translation accuracy, and ensuring authenticity of inscriptions (Gunarathna, 2020). The field thus is at the intersection of archaeology, linguistics, and artificial intelligence, striving to develop robust, interpretable artificial intelligence systems for epigraphic research.

1.2 Research Gap

Ancient Brahmi inscription analysis, especially those of the Anuradhapura period of Sri Lanka, is extremely important to preserve and fulfill the precious cultural and historical heritage hidden in these scripts. However, according to Hettiarachchi et al. (2020), there exists a broad gap between traditional archaeological approaches and modern computational methodologies. Archaeologists mainly rely on manual, slow, error-prone transcription and translation methods with no scalability. The absence of a computerized, strong, and easy-to-use system for translating Brahmi inscriptions to modern Sinhala script considerably hinders research speed and accuracy. The "Brahmi to Sinhala Translator" project addresses some of the fundamental concerns such as noise removal from scanned images, character recognition in the face of extensive script variability, and natural language processing of transliterations. However, existing solutions are fragmented and limited in nature, reflecting an overall research gap.

Additionally, the contemporary OCR techniques and AI-based recognition systems face tremendous difficulty due to the ancient, complex character shapes, script degradation, and the diverse ligature forms of Brahmi inscriptions (Chawla et al., 2024). Although deep learning models, particularly convolutional neural networks (CNNs), have proved to be promising with accuracy improvements in recognizing ancient scripts like Tamil and MODI (Chawla et al., 2024; Mahajan et al.), their application to Brahmi is nascent and limited. In addition, most of the models do not incorporate restoration feedback loops, a crucial element to handle the eroded and partial nature of archaeological texts effectively.

Within the field of semantic knowledge extraction, Graciotti (2023) demonstrates that although neural and symbolic methods can convert multi-lingual historical texts to knowledge graphs via state-of-the-art semantic parsing, such systems are usually tested on datasets and domains different from Brahmi inscriptions. Domain-specific ontologies, culturally aware lexicons, and end-to-end pipelines incorporating OCR, translation, named entity recognition (NER), topic modeling, and knowledge graph creation do not exist for ancient Sri Lankan epigraphy.

Furthermore, while knowledge graph-based recommendation systems are an area with a growing body of work (A, B, C), they are typically applied to modern, well-curated datasets such as scientific papers or generic social media posts and do not address the specific challenges of multilayered historical text analysis or restoration-based feedback for damaged texts. More critically, they lack multilingual translation modules and do not support deeper linguistic and cultural analyses required by Anuradhapura Brahmi inscriptions.

Therefore, the state of the art today has an evident gap: the absence of a monolithic, adaptive, and extensible AI-based system with the ability to accurately translate, semantically interpret, and iteratively enhance ancient Brahmi inscriptions via a restoration-aware feedback process. Bridging this gap will open an unprecedented opportunity to digitize, preserve, and expand our understanding of Sri Lankan historic texts, immensely benefiting archaeologists, linguists, historians, and the general cultural heritage society.

Feature / Capability	Paper A	Paper B	Paper C	Paper D	Paper E	Proposed System
OCR for ancient Brahmi script and preprocessing	✗	✗	✗	✓	✓	✓
Neural or hybrid translation model for ancient to modern language	✗	✗	✗	✗	✓	✓
NLP-driven named entity recognition and semantic extraction	✓	✗	✗	✗	✓	✓

Feature / Capability	Paper A	Paper B	Paper C	Paper D	Paper E	Proposed System
Integration of AI restoration feedback for damaged or incomplete text	X	X	X	X	X	✓
Interactive querying and graph-based visualization	✓	✓	✓	X	X	✓
Multilingual support and cultural heritage focus	X	X	X	X	✓	✓

Table 1 : Comparison between existing systems

Paper A

R. Manrique and O. Marino, “Knowledge graph-based weighting strategies for a scholarly paper recommendation scenario,” in *Proc. 12th Workshop on Knowledge-Aware and Conversational Recommender Systems*, 2018, pp. 1–8. [Online]. Available: https://ceur-ws.org/Vol-2290/kars2018_paper2.pdf

Paper B

Z. Ma, D. Wang, and J. Zhang, “Personalized scientific paper recommendation based on heterogeneous graph representation,” *Semantics Scholar*, 2018. [Online]. Available: <https://www.semanticscholar.org/paper/Personalized-Scientific-Paper-Recommendation-Based-Ma-Wang/92e3ea22703e5a5b8f5997d3781a79f23a923b7a>

Paper C

X. Wang et al., “Knowledge Graph Convolutional Networks for Recommender Systems,” in *Proc. 42nd Int. ACM SIGIR Conf. Research and Development in Information Retrieval*, 2019, pp. 1–10. [Online]. Available: <https://dl.acm.org/doi/10.1145/3331184.3331267>

Paper D

S. Ruwanmini, K. Dias, C. Niluckshini, and T. Nandasara, “Sinhala Inscription Character Recognition Model Using Deep Learning Technologies,” *Int. J. Adv. ICT Emerg. Regions*, vol. 16, no. 1, pp. 1–12, Mar. 2023. [Online]. Available: https://www.researchgate.net/publication/371922864_Sinhala_Inscription_Character_Recognition_Model_using_Deep_Learning_Technologies

Paper E

N. Hettiarachchi, I. Pathiraja, and D. Madawala, “Translation of Brahmi Inscriptions to Sinhala Using Natural Language Processing,” *Int. J. Comput. Appl.*, vol. 177, no. 30, pp. 19–25, Jan. 2020. [Online]. Available: <https://ijcaonline.org/archives/volume177/number30/hettiarachchi-2020-ijca-919776.pdf>

1.3 Research Problem

The central research question is the creation and deployment of an AI-augmented system that converts old, typically broken and damaged Brahmi inscriptions into modern Sinhala and English languages accurately and derives structured, semantically rich historical knowledge from the translations. This is compounded by a series of interrelated issues: physical wear on inscriptions such that characters become indistinct or missing; inflected morphological and syntactic features of the Brahmi and Old Sinhala scripts; ambiguity in linguistic sense combined with semantic change over time and geographical variation; and unavailability of large-scale, annotated training data that encompasses the lexicon and grammar of ancient inscriptions.

Addressing this problem involves an interdisciplinary research approach, combining state-of-the-art deep learning models for optical character recognition adjusted for image degradation in inscriptions, innovative NLP techniques such as Named Entity Recognition for identification of culturally and historically significant entities, topic modeling for thematic clusters, and knowledge graph construction for modeling relationships amongst discovered entities. Integrating these with feedback loops from completion/restoration modules filling gaps iteratively in mutilated texts is needed to augment translation coherence and semantic completeness.

Conquering these challenges effectively will enable converting scattered, deteriorated archaeological inscriptions into readable, trustworthy bilingual translations and semantic knowledge bases in-depth. This practice will benefit historians, archaeologists, linguists, and the masses by improving historical inquiry, digitalization of cultural heritage, interactive search of knowledge, and educational dissemination all key to maintaining and understanding Sri Lanka's rich Anuradhapura heritage.

2. OBJECTIVES

2.1 Main Objective

To develop an AI-driven, end-to-end system that would be able to precisely translate ancient Anuradhapura Brahmi inscriptions into modern Sinhala and English and unearth organized, semantically enriched historical, cultural, and administrative knowledge. The system will aid research by scholars, electronic documentation of Sri Lanka's ancient epigraphic heritage, and interactive discovery.

The central objective is to create a sophisticated computational platform that enables the transition between sensitive, partially decipherable ancient Brahmi inscriptions and modern digital humanities research. By means of AI-based translation and knowledge extraction, the system is designed to accomplish not just linguistic translation but contextual enrichment of inscriptions with inter-relations among entities, themes, and events. The result will enable historians, archaeologists, and cultural researchers to study, access, and disseminate ancient writings with unprecedented accuracy and scope, safeguarding Sri Lanka's valuable heritage.

2.2 Specific Objectives

- 1. Train and improve OCR models to recognize ancient Brahmi script on inscriptions that have grown old over time.**

Due to the physical deterioration and wear and tear of various inscriptions, this aim is focused on creating strong optical character recognition systems. These systems, as per deep learning design such as convolutional neural networks (CNNs), shall be trained on human-annotated images of Brahmi characters. The aim is to accurately segment and identify individual characters even in the case of incomplete and eroded or noisy inscriptions, thereby enabling strong digitization of scripts.

- 2. Create and implement natural language processing (NLP) based translation systems that convert recognized Brahmi characters into accurate Sinhala and English language words along with cultural and historical contexts.**

Ancient Brahmi language has some special vocabulary and grammatical structure not found in modern languages. The work involves combining rule-based linguistic theory with machine learning models of translation in order to deal with ambiguity and information gaps. The system will ensure the translations are meaningful based on archaeological evidence and are readable by scholars as well as the general public.

- 3. Apply NLP techniques like Named Entity Recognition (NER) and topic modeling to extract translated inscriptions and fetch useful historical and cultural entities.**

The objective is to enrich the translated inscriptions with automatically recognized and labeled important historical figures, places, religious words, and other useful concepts. By topic modeling to classify inscriptions in terms of topics, the research will reveal underlying patterns, facilitating easy understanding and classifying the epigraphic corpus.

- 4. Offer interactive knowledge graphs linking entities and topics that have been identified, enabling efficient querying and discovery of inscription data.**

After semantic extraction, the objective seeks to create knowledge graphs that model past personalities, geographical locations, occurrences, and relationships as nodes and edges. The system will facilitate researchers by providing a user-friendly and easy-to-use interface for visual and exploratory analysis of the interconnected data for the formulation of new insights and to facilitate heritage preservation operations.

5. Use an iterative feedback loop of image and text restoration using AI-driven algorithms to enhance the credibility of recognition and translation, especially for incomplete or defaced inscriptions.

Since most inscriptions are fragmentary or lack segments of text, the system will implement restoration algorithms that predict and fill-in missing text segments. Fill-in-the-blanks will be processed by the OCR and translation modules, so incremental improvements are achieved, and ever more complete and accurate outputs are provided to scholars.

6. Validate the accuracy and overall performance of the translation and knowledge extraction modules by expert validation and quantitative performance measures.

This objective will involve implementing testing procedures where experienced epigraphists in the ancient Brahmi language will test the system output for linguistic accuracy and cultural sensitivity. In pursuit of supporting expert judgment, measurable factors such as recognition correctness, entity extraction accuracy, and knowledge graph completeness will guide iterative improvement and verify the performance and scholarly significance of the system.

3. METHODOLOGY

3.1 Overall System Architecture

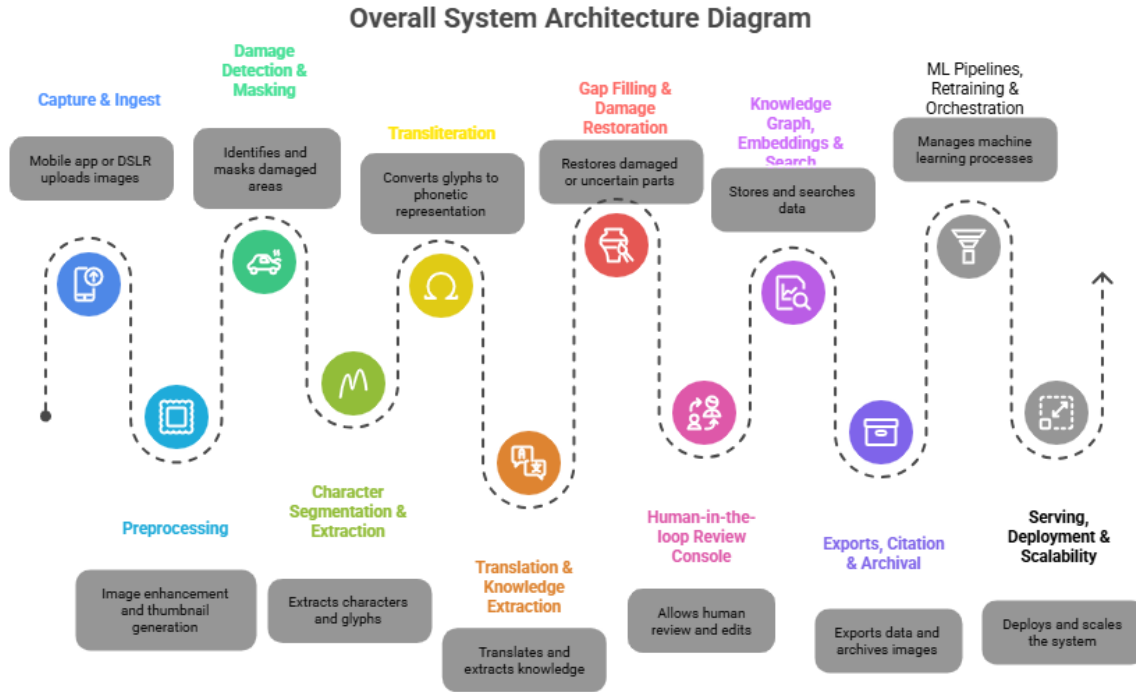


Figure 1 : Overall System Architecture Diagram

The workflow of ancient inscription processing begins from high-quality image acquisition of inscriptions with mobile phones or DSLR cameras. The images proceed through preprocessing steps such as removal of noise, stretching of contrast, and thumbnail generation to improve clarity and support background artefacts. Individual glyphs are then isolated from the inscription images by character segmentation and character extraction through projection-based or neural network methods. The segmented characters are subsequently transliterated into phonetic or symbolic forms through machine learning models and mapping rules. Gap-filling and damage recovery algorithms predict missing segments based on image and contextual features in order to restore damaged or missing components of the text. The clean and restored text is translated and semantically processed through natural language processing processes comprising named entity recognition and relationship extraction. Extracted entities and their relationships are organized into a

knowledge graph that is stored in a graph database for efficient querying and visualization. A human-in-the-loop review process is built in for facilitating expert feedback and correction, continuously refining the outputs. Export, archiving, and citation functionality is also part of the system, whereas machine learning pipelines allow for ongoing retraining and improving model performance. The whole pipeline is run on scalable infrastructure for supporting large volumes of data and multiple users reliably.

3.2 Component Diagram and overview

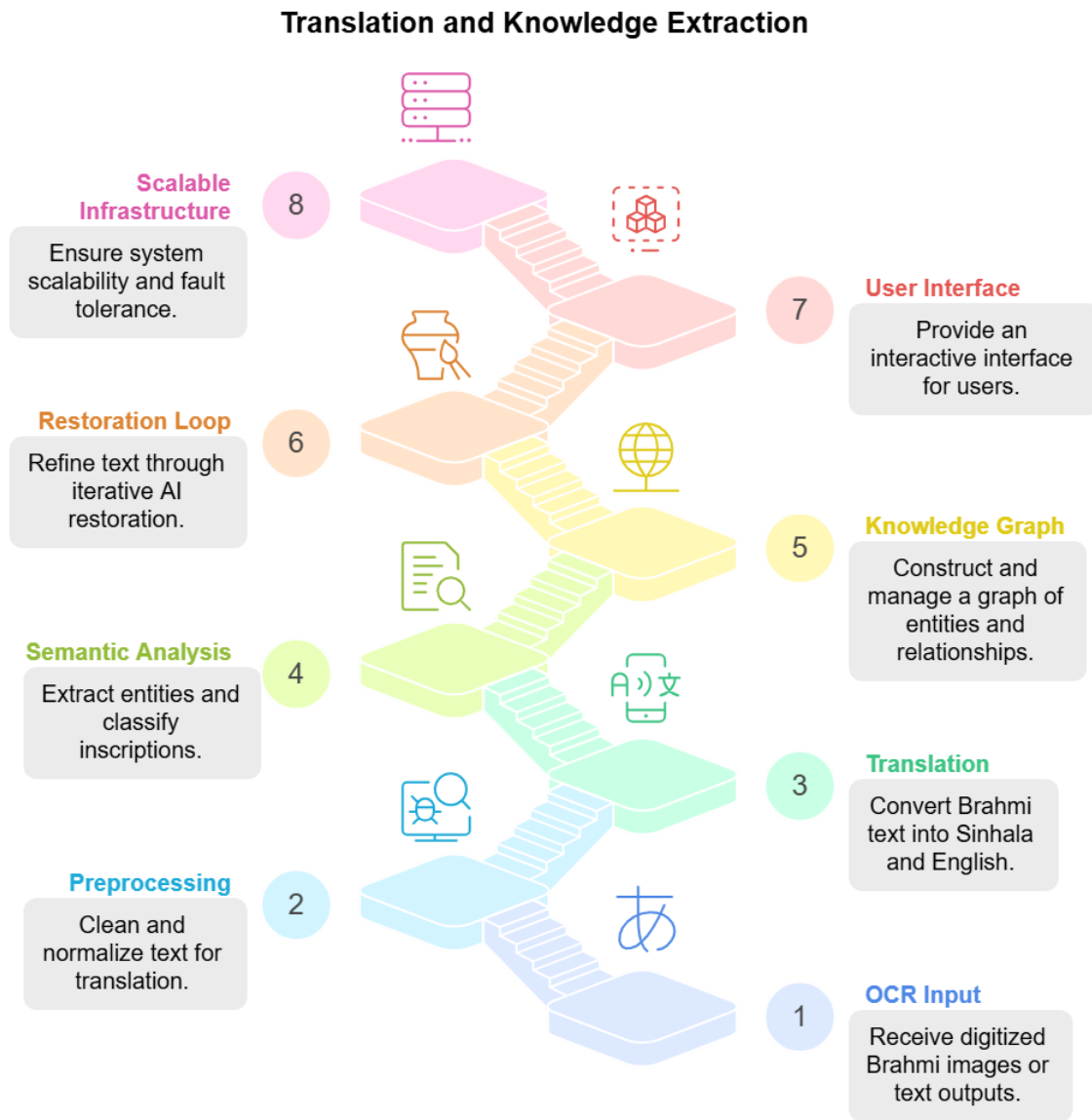


Figure 2: Overall Component Diagram

The proposed system for Component 3 is a comprehensive AI-powered platform that can translate ancient Brahmi inscriptions from digital texts into modern Sinhala and English and extract structured historical knowledge in a useable form for researchers. The system is a modular, iterative pipeline that has user inputs, AI processing modules, and databases that dynamically interact to ensure translation accuracy, semantic depth, and interactive discovery.

System and Flow of Information

1. User Interface (Mobile App / Web Page):

Researchers, epigraphists, and historians use the system via an easy-to-use mobile app or web page. Users can submit digitized image or Brahmi output text, request processing, and request translations and knowledge graphs. Graphical features allow semantic relations among historical objects from the inscriptions to be investigated.

2. Input Source (OCR Output / Digitized Text Repository):

The system accepts raw Brahmi script data from the OCR module (Component 2), which may contain uncertain or missing characters depending on the physical inscription damage. This digital text database is the primary source of input for translation as well as semantic analysis.

3. Preprocessing and Text Normalization Module:

This module cleanses, normalizes, and filters out the digitized Brahmi texts to handle OCR noise and inconsistency. It transforms variant spellings to standard ones and marks doubtful characters, presenting clean inputs for subsequent linguistic processing.

4. Machine Translation Module:

A rule-based linguistic mapping and trained machine learning-based hybrid translation engine translates the preprocessed Brahmi into Sinhala and English. The Contextual lexicons and archaic vocabulary and syntax rules for historical grammar rules are employed in this module to enhance translation accuracy.

5. Natural Language Processing (NLP) Semantic Analysis Module

Sophisticated NLP tools are applied here to improve translations. Named Entity Recognition (NER) detects significant historical figures, places, events, and terms; Topic Modeling classifies inscriptions by thematic content; Relationship Extraction determines semantic connections within and between the inscriptions.

6. Knowledge Graph Construction and Database

Extracted entities and relationships are structured into a semantic knowledge graph in RDF triples. The graph is stored and maintained in a Neo4j graph database to enable efficient query and interactive visualization within the user interface.

7. Iterative Restoration Feedback Loop:

To restore missing or destroyed text regions, restoration modules using AI (Component 4) read images/text to restore and suggest fixes. Restorations loop back to the preprocessing and translation modules, fine-tuning outputs with each iteration to improve accuracy and completeness.

The system delivers bilingual (Sinhala and English) translations as well as interactive visualizations of knowledge graphs via the user interface. This allows researchers to browse through the connected semantic knowledge and gain more extensive historical information. This system design harmoniously integrates various AI modules, user interaction layers, and knowledge management systems to create a stable, scalable platform for digitalization and semantic enrichment of ancient Brahmi inscriptions.

4. PROJECT REQUIREMENTS

4.1 Functional Requirements

1. Input Text Processing

The system will accept digitized Brahmi script text output from the prior OCR module, including doubtful or incomplete characters from shattered inscriptions, and perform preprocessing, normalization, and noise removal to prepare the data for translation and examination.

2. Bilingual Translation

The system will translate Brahmi text into both Sinhala and English with linguistic accuracy and cultural suitability, employing hybrid rule-based and machine learning-based translation architectures.

3. Semantic Knowledge Extraction

The system will employ natural language processing techniques such as Named Entity Recognition (NER), topic modeling, and relationship extraction to identify and classify historical persons, places, events, and thematic content in translated texts.

4. Iterative Restoration Feedback Integration

The system will incorporate AI-based restoration model outputs for iteratively fine-tuning and improving translational precision and knowledge extraction through correcting missing or erased text sections.

4.2 Non-Functional Requirements

1. Security

The system will incorporate user authentication and role-based access control to protect sensitive scholarly and archaeological information.

2. Scalability

The architecture will scale without loss of system performance to accommodate growing amounts of inscription data, user queries, and semantic knowledge graph complexity.

3. Usability

The user interfaces will be intuitive and accessible to enable researchers with varying technical ability to navigate, query, and interpret translation and knowledge graph results effectively.

4. Availability

The system will ensure maximum uptime and redundancy mechanisms to offer uninterruptable service to research and cultural heritage preservation activities.

5. Performance

The system will ensure fast response for translation, semantic extraction, and graph querying operations, with the goal of minimal latency (e.g., less than 10 seconds per inscription processing).

4.3 System Requirements

- Development Platforms & Tools
 1. Python programming language
 2. Deep learning frameworks like TensorFlow or PyTorch for OCR, NLP, and restoration models
 3. NLP libraries like SpaCy or NLTK for semantic processing
 4. Neo4j graph database for knowledge graph storage and querying, using RDF triples for standardized semantic structure
 5. Visualization frameworks capable of handling interactive graph interfaces
 6. Agile development methodology to enable iterative feature implementation and integration
- Hardware & Infrastructure
 1. High-end GPU-enabled machines or cloud services for deep learning training (OCR, translation, restoration)
 2. Cloud-scalable servers for hosting graph databases and web portals
- Data Resources
 1. Brahmi inscription datasets with annotated data for OCR and NLP training
 2. Historical and linguistic dictionaries of Sinhala, Brahmi, and English
 3. Restoration annotated datasets for iterative feedback loops

4.4 User Requirements

1. Researchers / Epigraphists
 - Need accurate bilingual translations with contextual notes appended
 - Want to query structured historical knowledge bases
 - Expect interactive visualization and theme exploration capabilities
2. Historians / Cultural Heritage Professionals
 - Desire actionable insights from translations and knowledge graphs to aid in heritage preservation
 - Require reliable data outputs validated by expert review
3. Project Technical Team
 - Need modular, maintainable system architecture to update components
 - Require logging and metrics for live performance and error monitoring
4. Educators and Public Users
 - Wish clear, understandable translations and accessible educational visualizations to increase cultural awareness

4.5 Expected Test Cases

Test Case ID	Description	Input	Expected Output
TC001	Preprocess digitized Brahmi text input	Raw OCR output with noisy, incomplete Brahmi characters	Cleaned, normalized Brahmi text ready for translation
TC002	Handle translation input with damaged or ambiguous text	Brahmi text containing missing/unclear characters	Best-effort translation with restoration feedback integrated
TC003	Perform Named Entity Recognition (NER) on translated texts	Translated text containing historical names, places, events	Correctly tagged entities with type labels (Person, Place, etc.)
TC004	Construct and populate knowledge graph from extracted data	Set of extracted entities and relations	RDF triples correctly representing relationships; graph stored in Neo4j
TC005	Interactive querying of knowledge graph	Query for specific entity or relation	Relevant entity nodes and relations displayed
TC006	Integrate restoration feedback for damaged inscriptions	Inputs with partial/missing text segments	Iteratively refined translations with increased completeness
TC007	User interface usability test for querying and visualizations	Researcher using knowledge graph UI	User successfully navigates, queries, and interprets data
TC008	Security testing for access control and data protection	Unauthorized and authorized access attempts	Unauthorized access denied, authorized users can access data
TC009	System performance and latency under load	Simulated multiple concurrent users accessing querying services	System remains responsive with latency within acceptable limits

TC010	Logging and error handling	Introduced errors or unexpected inputs	Errors logged correctly, system recovers or fails gracefully
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Table 2: Test Cases

5. BUDGET & BUDGET JUSTIFICATION

Requirement	Cost (Rs.)
Fieldwork/Travel for Data Collection	Rs. 25,000.00
Purchasing virtual storage	Rs. 4,000.00
Software Licenses (OCR/Image Tools)	Rs. 6,000.00
Miscellaneous (app store fees etc.)	Rs. 5,000.00
Hosting	Rs. 5000.00 per month
Total Cost	Rs. 45,000.00 + monthly hosting fee

Table 3: Budget Details

Justifications:

- Fieldwork/Travel for Data Capturing (Rs. 25,000.00)

This is for visiting sites and capturing real images of inscriptions. It covers transportation, travel, and modest expenses during data capturing.

- Purchasing Virtual Storage (Rs. 4,000.00)

Since images take up a lot of space, more space is needed to safely store all the data and make it readily available to members of the team.

- Software Licenses (OCR/Image Tools) (Rs. 6,000.00)

Special software will be required for image and text processing. The license cost will allow us to make use of advanced features for accurate results.

- Miscellaneous (Rs. 5,000.00)

This is kept for small or unexpected costs such as app store fees or other minor necessities that might crop up.

- Hosting (Rs. 5,000.00/month)

The system should be hosted online so that it is readily accessible. The monthly subscription will maintain the system operational and always in service.

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